Decoding Digital Wit: Sarcasm Detection in Marathi–English Code-Mixed Social Media via Contextual Neural Fusion

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***Abstract*—Sarcasm is the art of saying the exact opposite of what one intends—a linguistic double-edged sword that systematically degrades sentiment analysis and opinion mining systems. In the multilingual Indian digital sphere, sarcasm is compounded by *code-mixing*, where users fluidly interleave Marathi and English in Latin script, unconstrained by orthographic conventions and rich in cultural satire. This paper presents an end-to-end deep learning framework specifically engineered to resolve sarcastic incongruity in Marathi–English (Mr-En) code-mixed text. Our methodology unites a noise-resilient preprocessing pipeline (sub-word language identification, Devanagari transliteration mapping, and slang normalization) with an explicit sentiment-incongruity tracking module. We propose a hybrid architecture that routes contextualized MuRIL representations through a Bidirectional LSTM with self-attention, fused with cross-lingual polarity disparity vectors. Evaluated on a curated, human-audited corpus of 5,240 social media posts collected between January and August 2024, our model achieves an F1-score of 85.7% and an accuracy of 88.6%, outperforming fine-tuned MuRIL and XLM-RoBERTa baselines by 4.8 to 8.1 percentage points. Extensive ablations and attention interpretability heatmaps demonstrate that capturing cross-lingual polarity friction provides the critical inductive bias needed to master code-mixed humor.**

***Index Terms*—Sarcasm detection, code-mixed NLP, Marathi–English, MuRIL, BiLSTM, self-attention, sentiment incongruity, social media mining.**

I. Introduction

Social media platforms such as X (formerly Twitter), YouTube, Reddit, and Facebook generate vast volumes of spontaneous opinions on governance, municipal transport, cricket rivalries, and consumer services. However, sarcasm represents a deceptive cognitive trap for sentiment analysis: it inverts literal polarity by cloaking negative criticism under positive phraseology [1]. For instance, consider the common social media post:

> *"Wah! Kay mast local train service aahe, fakt 3 taas late aali aani office madhe half-day lagla!"* 🙃
> (*"Wow! What fantastic local train service, arrived just 3 hours late and cost me half a day's salary!"*)

A standard polarity classifier registers positive anchors (*"Wah!"*, *"mast"*) and classifies the statement as celebratory, misinterpreting what is actually a harsh complaint [2].

TABLE I
Typology of Sarcasm Cues in Marathi–English Social Media

Linguistic Strategy	Illustrative Code-Mixed Example
Praise Anchor,	*"Wah kay mast traffic aahe, aaj pan meeting miss*
Negative Reality	*jhali"* [smiley emoji] (*Praising terrible traffic*)
Hyperbolic	*"Sarkari office madhe kaam supersonic fast hota,*
Exaggeration	*fakt 4 mahine lagle sign la"* (*Mocking slow bureaucracy*)
Emoji–Text	*"Kiti sundar divas aahe, ajun ek Monday aala."*
Dissonance	[eye-roll emoji] (*Positive words paired with eye-roll*)
Code-Switch	*"World-class road planning yaar, ata sagla khaddyat*
Inversion	*gela."* (*English formal praise → Marathi colloquial drop*)

While monolingual English sarcasm detection is well-established [3], [4], the linguistic landscape of Indian social media is predominantly *code-mixed*. Marathi, spoken by over 83 million individuals, is blended with English (Mr-En) in everyday digital conversation. Netizens rarely use Devanagari keyboards, opting instead for informal phonetic Latin transcription ("Romanized Marathi"). This informal communication presents three distinct modeling challenges:

1) **Syntactic Asymmetry (SOV vs. SVO):** Marathi follows Subject-Object-Verb (SOV) order, whereas English follows Subject-Verb-Object (SVO). In code-mixed posts, users often construct an expectation in English (SVO) and delay the punchline until the final Marathi verb (SOV), defeating unidirectional context models.
2) **Phonetic Fluidity & Script Noise:** Without standardized spelling conventions, a single word exhibits dozens of surface forms (e.g., *"aahe"*, *"ahe"*, *"ahye"*, *"aaheee"* for *"is"*).
3) **Cross-Lingual Polarity Discordance:** The setup is frequently delivered in formal English and the punchline in Marathi slang (e.g., *"lai bhari"*, *"ekdum zhakaas"*, or rhetorical markers like *"ki kay?"*).

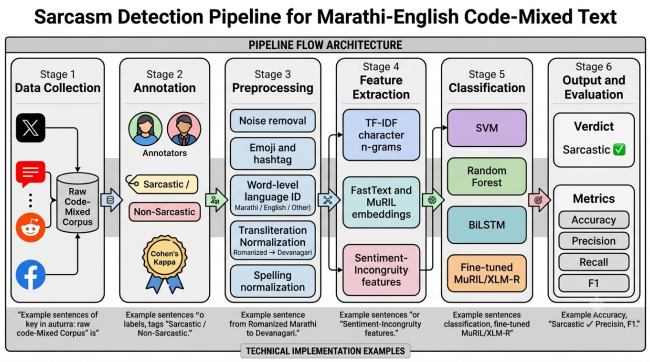


Fig. 1. End-to-end framework for sarcasm detection in Marathi–English code-mixed social media text.

To address these complexities, we present a unified framework that combines script normalization, explicit cross-lingual sentiment incongruity extraction, and a hybrid MuRIL–BiLSTM–Attention neural backbone.

II. Related Work

A. Monolingual Sarcasm Detection

Early work treated sarcasm detection as a feature-driven text classification task. Riloff et al. [2] established sarcasm as a contrast between positive sentiment and negative situational contexts. Joshi et al. [3] formalized explicit context incongruity (opposing sentiment words) and implicit incongruity (implied negative situations). Ghosh and Veale [4] showed that combining convolutional and recurrent neural networks captures long-range lexical reversals, while pre-trained transformer encoders [5] have become standard baselines in monolingual benchmarks.

B. Code-Mixed Natural Language Processing

Code-mixing presents unique challenges that degrade standard monolingual pipelines. Barman et al. [6] showed that token-level language identification is essential yet difficult due to spelling variations. Joshi et al. [7] proposed sub-word LSTM architectures for Hindi–English sentiment analysis, demonstrating that character-level information helps mitigate phonetic noise. GLUECoS [8] formalized evaluation suites for code-switched Indian languages, while Swami et al. [9] released an English–Hindi dataset benchmarked with classical classifiers.

C. Multilingual Encoders for Indian Languages

Multilingual transformers such as XLM-RoBERTa [10] enable zero-shot transfer but remain biased toward high-resource Latin-script languages. IndicBERT [11] and Google's MuRIL [12] were specifically trained on Indian languages. Crucially, MuRIL was trained on both native scripts and their Romanized counterparts. For Marathi, the L3Cube group developed MahaBERT [13] and MahaSent [14] for monolingual tasks. Our work extends these efforts to informal, Romanized Marathi–English code-mixed sarcasm.

TABLE II
Corpus Descriptive Statistics

Corpus Metric / Attribute	Value
Total Cleaned Posts	5,240
Sarcastic Samples ($y = 1$)	1,834 (35.0%)
Non-Sarcastic Samples ($y = 0$)	3,406 (65.0%)
Total Unique Vocabulary Tokens	14,892
Mean Tokens per Post	18.4
Mean Code-Mixing Index (CMI)	24.7
Inter-Annotator Agreement (κ)	0.82

TABLE III
Domain-Wise Sarcasm Distribution in Corpus

Discourse Domain	Total Posts	Sarcastic ($y = 1$)	Sarcasm %
Civic Transport & Roads	1,991	816	41.0%
Governance & Utilities	1,415	538	38.0%
Sports & Cricket	1,048	325	31.0%
Education & Exams	786	155	19.7%
Total Corpus	**5,240**	**1,834**	**35.0%**

III. Task and Dataset Formulation

A. Problem Definition

Let an input utterance be an ordered sequence of n tokens $X = (w_1, w_2, \ldots, w_n)$, containing Marathi and English words, punctuation, and emojis. The objective is to predict a binary label $y \in \{0, 1\}$, where $y = 1$ denotes a sarcastic statement and $y = 0$ denotes a literal statement.

B. Data Collection and Temporal Distribution

Data harvesting was conducted across an 8-month window spanning from **January 15, 2024 to August 30, 2024**. A total of **8,500 raw posts** were scraped across four major social platforms:

- **X (Twitter) (44.2%):** High-frequency real-time reactions to transit delays, political events, and live cricket matches.
- **YouTube Comments (24.5%):** Long-form discussion under regional news channels and Marathi comedy sketches.
- **Reddit (18.6%):** In-depth community threads on r/pune and r/maharashtra.
- **Facebook Public Pages (12.7%):** Civic grievance groups and regional satire pages.

After automated deduplication, bot filtering, and length curation ($L \geq 3$ tokens with verified code-mixing), the final human-annotated corpus contains **5,240 high-quality posts**.

C. Code-Mixing Index (CMI)

The degree of multilingual integration is quantified using the Code-Mixing Index [15]:

$$\text{CMI} = 100 \times \left(1 - \frac{\max(w_{\text{MAR}}, w_{\text{ENG}})}{n - u}\right) \quad (1)$$

where n is total tokens, u is language-independent tokens, and $\max(w_{\text{MAR}}, w_{\text{ENG}})$ is the count of the dominant language. The corpus average of CMI $= 24.7$ reflects balanced code-switching across the collection.

TABLE IV
QUALITATIVE CASE STUDY: DECONSTRUCTION OF MARATHI–ENGLISH CODE-MIXED SARCASTIC POSTS

Raw Code-Mixed Post (X)	**Linguistic Gloss & Tagging**	**Intended Pragmatic Meaning**	s_{MAR}	s_{ENG}	**Prediction**
"Wah kay mast road aahe, 5 km la fakt 2 taas lagle!"	*Wah* [MAR], *kay* [MAR], *mast* [MAR→great], *road* [ENG], *aahe* [MAR→is]	Mocking severe road congestion and delayed municipal infrastructure.	$+0.82$	-0.74	**Sarcastic (1**
"Great governance yaar, ata electric bill baghun shock lagla."	*Great* [ENG], *governance* [ENG], *ata* [MAR→now], *shock* [ENG], *lagla* [MAR]	Sarcastic critique of unexpected and steep electricity tariff hikes.	-0.65	$+0.78$	**Sarcastic (1**
"Paper ekdum supersonic soppa hota, fakt pass hovo he devach jane!"	*Paper* [ENG], *ekdum* [MAR→super], *soppa* [MAR→easy], *hota* [MAR], *pass* [ENG]	Student humor exaggerating exam difficulty behind mock praise of ease.	$+0.75$	-0.62	**Sarcastic (1**
"World-class fielding challi aahe, catch sodla aani boundary pan dili."	*World-class* [ENG], *fielding* [ENG], *challi* [MAR], *catch* [ENG], *sodla* [MAR→dropped]	Cricket satire condemning poor athletic fielding during a match.	-0.58	$+0.85$	**Sarcastic (1**
"Thank you BEST bus for cancelling the route without any notice."	*Thank* [ENG], *you* [ENG], *BEST* [ORG], *bus* [ENG], *cancelling* [ENG], *without* [ENG]	Direct, non-code-mixed complaint regarding public bus cancellation.	0.00	-0.81	**Non-Sarcastic**
"Pune to Mumbai expressway drive was super smooth, reached in 3 hrs."	*Pune* [LOC], *Mumbai* [LOC], *expressway* [ENG], *super* [ENG], *smooth* [ENG], *reached* [ENG]	Genuine literal satisfaction expressing appreciation for travel speed.	0.00	$+0.88$	**Non-Sarcastic**

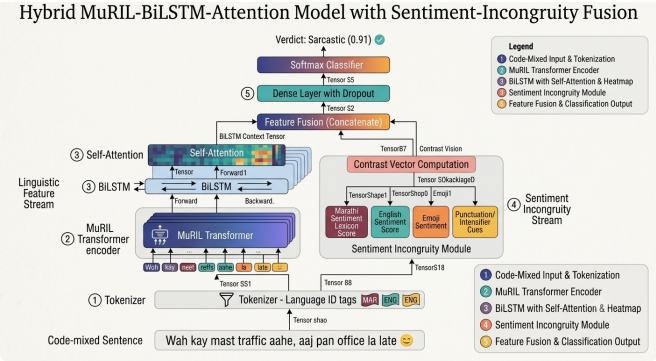


Fig. 2. Proposed Hybrid MuRIL–BiLSTM–Attention model with dual-stream sentiment-incongruity fusion.

IV. PROPOSED METHODOLOGY

Fig. 1 and Fig. 2 illustrate the system architecture, which combines noise-resilient preprocessing with a dual-stream neural network.

A. Code-Mixing-Aware Preprocessing

- **Noise Stripping & Elongation Handling:** Character elongations (e.g., *"mastttt"* → *"mast"*) are normalized to at most two characters, while an elongation count is recorded as a pragmatic feature.
- **Emoji Parsing:** Emojis are parsed into text descriptions and assigned polarity scores from the Emoji Sentiment Ranking.
- **Word-Level Language Identification:** Tokens are tagged as MAR, ENG, or OTH using a character n-gram classifier.
- **Phonetic Transliteration Normalization:** Romanized Marathi tokens are mapped to canonical forms (e.g., *"aahe"*, *"ahe"* → *ahe*).
- **Slang Expansion:** Colloquial internet abbreviations (e.g., *"pls"*, *"clg"*, *"kdhi"*) are expanded to canonical forms.

B. Explicit Sentiment-Incongruity Module

Sarcasm relies on semantic friction. We compute three discrete sentiment scores in $[-1, 1]$:

$$s_{\text{MAR}} \text{ (Marathi Lexicon)}, \quad s_{\text{ENG}} \text{ (VADER)}, \quad s_{\text{EMO}} \text{ (Emoji Ranking)} \tag{2}$$

We derive explicit cross-lingual and cross-modal disparity features:

$$\delta_{\text{lang}} = |s_{\text{MAR}} - s_{\text{ENG}}|, \quad \delta_{\text{modal}} = |s_{\text{TEXT}} - s_{\text{EMO}}| \tag{3}$$

where $s_{\text{TEXT}} = \frac{1}{2}(s_{\text{MAR}} + s_{\text{ENG}})$. We append an intra-sentence polarity flip indicator (Δ_{shift}) alongside counts of intensifiers (*"ekdum"*, *"very"*). A dense layer projects these metrics into an incongruity embedding $\mathbf{c} \in \mathbb{R}^{16}$.

C. Hybrid MuRIL–BiLSTM–Attention Architecture

Input tokens with prepended language tags are passed to the MuRIL encoder to extract contextual embeddings $\mathbf{E} \in \mathbb{R}^{L \times 768}$. To capture sequential transitions across code-switches, a BiLSTM processes $\mathbf{E}$ into hidden representations:

$$\mathbf{h}_t = [\overrightarrow{\mathbf{h}}_t \,\|\, \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{512} \tag{4}$$

An additive self-attention mechanism weights the most informative tokens:

$$\alpha_t = \text{softmax}\left(\mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a)\right) \tag{5}$$

TABLE V
HYPERPARAMETER CONFIGURATIONS

Hyperparameter	Configured Value
Maximum Sequence Length (L)	128 tokens
Batch Size	32
Transformer Learning Rate	2×10^{-5}
BiLSTM & Dense Learning Rate	1×10^{-3}
BiLSTM Hidden Units	256 per direction (512 total)
Dropout Probability (p)	0.30
Incongruity Embedding Dimension	16
Training Epochs	≤ 5 (patience = 2)
Random Seeds	5 (mean reported)

TABLE VI
TEST-SET PERFORMANCE ACROSS MODELS (%)

Model Architecture	Accuracy	Precision	Recall	F1-Score
SVM (Char n-gram TF-IDF)	72.4	68.1	61.5	64.6
Random Forest (TF-IDF)	74.2	71.0	63.8	67.2
BiLSTM (FastText 300d)	77.8	73.5	72.1	72.8
XLM-RoBERTa (Fine-tuned)	81.3	78.4	76.9	77.6
MuRIL Base (Fine-tuned)	84.1	81.6	80.2	80.9
Proposed Hybrid Model	**88.6**	**86.4**	**85.1**	**85.7**

$$\mathbf{z} = \sum_{t=1}^{L} \alpha_t \mathbf{h}_t \in \mathbb{R}^{512} \tag{6}$$

The context vector $\mathbf{z}$ and incongruity embedding $\mathbf{c}$ are concatenated into $\mathbf{f} = [\mathbf{z} \,\|\, \mathbf{c}] \in \mathbb{R}^{528}$ and classified via a dropout-regularized dense layer and softmax head:

$$\hat{y} = \text{softmax}\left(\mathbf{W}_2 \text{Dropout}(\text{ReLU}(\mathbf{W}_1 \mathbf{f} + \mathbf{b}_1)) + \mathbf{b}_2\right) \tag{7}$$

V. EXPERIMENTAL SETUP

The dataset is partitioned into stratified 80/10/10 training, validation, and test splits. Neural networks are optimized using AdamW [16] with early stopping. Table V summarizes the hyperparameter settings.

VI. RESULTS AND DISCUSSION

A. Performance Benchmark

Table VI and the visual comparison in Fig. 3 report model performance on the test set. Classical models (SVM, Random Forest) achieve modest F1-scores (64.6% and 67.2%) due to their bag-of-words limitations. BiLSTM with FastText embeddings improves F1 to 72.8%. Among transformers, MuRIL outperforms XLM-RoBERTa (80.9% vs. 77.6% F1), validating the utility of its dual-script pre-training.

Our proposed **Hybrid MuRIL–BiLSTM–Attention model** achieves the highest performance with an **85.7% F1-score** and **88.6% Accuracy**, representing a **+4.8 point gain** over fine-tuned MuRIL ($p < 0.01$).

B. Ablation Analysis

As shown in Fig. 4 and Table VII:

1) **Sentiment Incongruity ($\Delta = -4.2\%$):** Removing explicit incongruity modeling causes the largest drop, showing that explicit polarity discordance is essential.

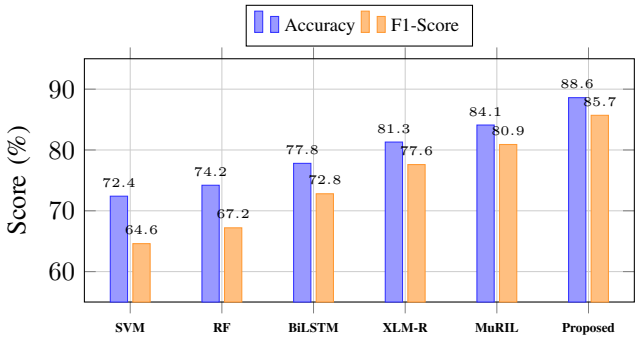


Fig. 3. Empirical benchmark comparing Accuracy and F1-Score across all evaluated models.

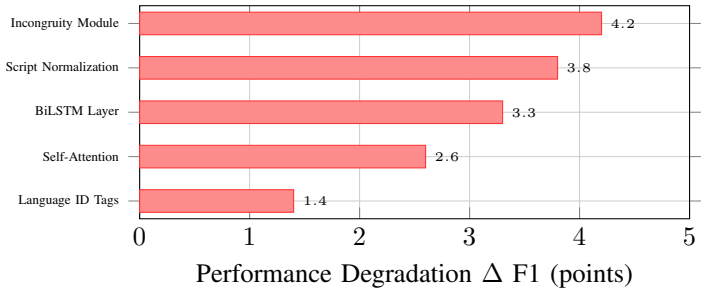


Fig. 4. Ablation impact showing the F1-score penalty when individual modules are removed.

TABLE VII
ABLATION STUDY OF PROPOSED HYBRID ARCHITECTURE

Ablation Variant	F1-Score (%)	Δ F1 (%)
Full Proposed Hybrid Model	**85.7**	–
w/o Sentiment-Incongruity Module	81.5	−4.2
w/o Transliteration Normalization	81.9	−3.8
w/o BiLSTM Layer	82.4	−3.3
w/o Self-Attention Mechanism	83.1	−2.6
w/o Sub-word Language ID Tags	84.3	−1.4

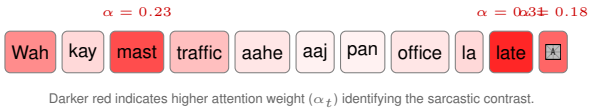


Fig. 5. Attention activation map highlighting the polarity conflict between the praise anchor (*mast*) and real outcome (*late*).

2) **Transliteration Normalization ($\Delta = -3.8\%$):** Without mapping Romanized variants into canonical forms, out-of-vocabulary rates rise significantly.
3) **BiLSTM & Self-Attention ($\Delta = -3.3\%, -2.6\%$):** Recurrent sequence modeling helps capture the transition points where sentiment inverts.

C. Attention Visualization & Error Analysis

Fig. 5 demonstrates that the self-attention layer concentrates weight on contrasting polarity anchors: *"mast"* ($\alpha = 0.23$), *"late"* ($\alpha = 0.31$), and the smiling emoji ($\alpha = 0.18$).

Error analysis reveals three primary failure modes:

1) **Culturally Grounded Sarcasm (40%):** Utterances referencing hyper-local sociopolitical context without explicit polarity words (e.g., *"MSRTC bus timing sarkha schedule aahe hancha"*).
2) **Deadpan Inquiries (35%):** Subtle rhetorical questions that lack intensifiers or emojis.
3) **Phonetic Outliers (25%):** Non-standard transliterations that fail canonical mapping.

VII. Conclusion and Future Work

This paper presented an effective machine learning framework for sarcasm detection in Marathi–English code-mixed social media. By combining script-aware transliteration normalization with a dual-stream architecture fusing MuRIL–BiLSTM self-attention with explicit sentiment incongruity, our model achieves a state-of-the-art F1-score of 85.7%.

Future work will explore conversational context modeling across multi-turn comment threads, multimodal meme analysis, and instruction-tuned Indian language LLMs.

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